$$S(g) = \int_M R \, dV_g \qquad dV_g = \sqrt{g} \, dx^0 \wedge \cdots \wedge dx^n$$

Let M be a smooth manifold and $g \in \Gamma(\Sigma^2 T^* M)$ s.t. $g_p : T_pM \times T_pM \to \mathbb{R}$ is nondegenerate and has $\underbrace{\text{signature } (1, n-1)}$

there exists a basis $v, u_1, \ldots, u_{n-1}$ s.t.

$$g_p(v,v) = -1$$
$$g_p(u_i, u_j) = \delta_{ij}$$
$$g_p(v, u_i) = 0$$

g is called Lorentzian metric.

Aside: In Minkowski geometry (special relativity) we have

$$\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$$

In Lorentzian geometry the volume form is given by $dV_g = \sqrt{-g} \, dx^0 \wedge dx^1 \wedge \cdots \wedge dx^n$

If $F \in \Gamma(T^{(k,l)}M)$ then $F = F^{j_1 \cdots j_l}_{i_1 \cdots i_k} \, dx^{i_1} \otimes \cdots \otimes dx^{i_k} \otimes \partial_{j_1} \otimes \cdots \otimes \partial_{j_l}$

$$\nabla F \in \Gamma(T^{(k+1,l)}M) \Rightarrow \nabla F = (\nabla F)^{j_1 \cdots j_l}_{i_1 \cdots i_k m} \, dx^{i_1} \otimes \cdots \otimes dx^{i_k} \otimes dx^m \otimes \partial_{j_1} \otimes \cdots \otimes \partial_{j_l}$$

$$(\nabla F)^{j_1 \cdots j_k}_{i_1 \cdots i_k m} = \partial_m F^{j_1 \cdots j_l}_{i_1 \cdots i_k} + \sum_{s=1}^{k} F^{j_1 \cdots \alpha \cdots j_l}_{i_1 \cdots i_k} \Gamma^{i_s}_{m\alpha} - \sum_{s=1}^{l} F^{j_1 \cdots j_l}_{i_1 \cdots \alpha \cdots j_k} \Gamma^{\alpha}_{m j_s}$$

$$S(g) = \int_M R \sqrt{-g} \, dx^0 \wedge \cdots \wedge dx^n$$

$$0 = \delta(\delta^\mu{}_\nu)$$
$$= \delta(g^{\mu\alpha} g_{\alpha\nu})$$
$$= g^{\mu\alpha} \delta g_{\alpha\nu} + g_{\alpha\nu} \delta g^{\mu\alpha}$$

$$\Rightarrow g^{\mu\alpha} \delta g_{\alpha\nu} = -g_{\alpha\nu} \delta g^{\mu\alpha}$$

$$\Rightarrow \delta g_{\beta\nu} = -g_{\mu\beta} g_{\alpha\nu} \delta g^{\mu\alpha}$$

$$D(\det)(A)(X) = (\det A)\,\mathrm{tr}(A^{-1}X)$$

$$\delta(-g) = (-g)\,g^{\mu\nu}\,\delta g_{\mu\nu}$$
$$= -(-g)\,g_{\alpha\beta}\,\delta g^{\alpha\beta}$$

Hence $\delta(\sqrt{-g}) = \frac{1}{2}(-g)^{-1/2}\,\delta(-g) = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\,\delta g^{\mu\nu}$

Next note $R = (\mathrm{Ric})^{\mu}{}_{\mu} = R^{\mu}{}_{\mu}$

we will show $\delta R_{\sigma\nu} = \nabla_\rho \delta\Gamma^\rho_{\nu\sigma} - \nabla_\nu \delta\Gamma^\rho_{\rho\nu}$ where $\nabla_\sigma = \nabla_{\partial_\sigma}$ [Palantini identity]

Recall $R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\mu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$

$$\Rightarrow \delta R^\rho{}_{\sigma\mu\nu} = \partial_\mu \delta\Gamma^\rho_{\mu\sigma} - \partial_\nu \delta\Gamma^\rho_{\mu\sigma} + (\delta\Gamma^\rho_{\mu\lambda})\Gamma^\lambda_{\nu\sigma} + \Gamma^\rho_{\mu\lambda}(\delta\Gamma^\lambda_{\nu\sigma}) - (\delta\Gamma^\rho_{\nu\lambda})\Gamma^\lambda_{\mu\sigma} - \Gamma^\rho_{\nu\lambda}(\delta\Gamma^\lambda_{\mu\sigma})$$

$$\nabla_\mu \delta\Gamma^\rho_{\nu\sigma} = \partial_\mu \delta\Gamma^\rho_{\nu\sigma} + \Gamma^\rho_{\mu\lambda}\delta\Gamma^\lambda_{\nu\sigma} - \Gamma^\lambda_{\mu\nu}\delta\Gamma^\rho_{\lambda\sigma} - \Gamma^\lambda_{\mu\sigma}\delta\Gamma^\rho_{\nu\lambda}$$

and so
$$\left.\begin{aligned} \partial_\mu \delta\Gamma^\rho_{\nu\sigma} &= \nabla_\mu \delta\Gamma^\rho_{\nu\sigma} - \Gamma^\rho_{\mu\lambda}\delta\Gamma^\lambda_{\nu\sigma} + \Gamma^\lambda_{\mu\nu}\delta\Gamma^\rho_{\lambda\sigma} + \Gamma^\lambda_{\mu\sigma}\delta\Gamma^\rho_{\nu\lambda} \\ \partial_\nu \delta\Gamma^\rho_{\mu\sigma} &= \nabla_\nu \delta\Gamma^\rho_{\mu\sigma} - \Gamma^\rho_{\nu\lambda}\delta\Gamma^\lambda_{\mu\sigma} + \Gamma^\lambda_{\nu\mu}\delta\Gamma^\rho_{\lambda\sigma} + \Gamma^\lambda_{\nu\sigma}\delta\Gamma^\rho_{\nu\mu} \end{aligned}\right\}$$
make assumption that torsion $=0$ hence $\Gamma^\alpha_{\beta\nu} = \Gamma^\alpha_{\nu\beta}$

$$\Rightarrow \partial_\mu \delta\Gamma^\rho_{\nu\sigma} - \partial_\nu \delta\Gamma^\rho_{\mu\sigma} + (\delta\Gamma^\rho_{\mu\lambda})\Gamma^\lambda_{\nu\sigma} + \Gamma^\rho_{\mu\lambda}(\delta\Gamma^\lambda_{\nu\sigma}) - (\delta\Gamma^\rho_{\nu\lambda})\Gamma^\lambda_{\mu\sigma} - \Gamma^\rho_{\nu\lambda}(\delta\Gamma^\lambda_{\mu\sigma})$$
$$= \nabla_\mu \delta\Gamma^\rho_{\nu\sigma} - \nabla_\nu \delta\Gamma^\rho_{\mu\sigma} - \cancel{\Gamma^\rho_{\mu\lambda}\delta\Gamma^\lambda_{\nu\sigma}} + \cancel{\Gamma^\lambda_{\mu\nu}\delta\Gamma^\rho_{\lambda\sigma}} + \cancel{\Gamma^\lambda_{\mu\sigma}\delta\Gamma^\rho_{\nu\lambda}} + \cancel{\Gamma^\rho_{\nu\lambda}\delta\Gamma^\lambda_{\mu\sigma}} - \cancel{\Gamma^\lambda_{\nu\mu}\delta\Gamma^\rho_{\lambda\sigma}} - \cancel{\Gamma^\lambda_{\nu\sigma}\delta\Gamma^\rho_{\nu\mu}}$$
$$+ \cancel{(\delta\Gamma^\rho_{\mu\lambda})\Gamma^\lambda_{\nu\sigma}} + \cancel{\Gamma^\rho_{\mu\lambda}(\delta\Gamma^\lambda_{\nu\sigma})} - \cancel{(\delta\Gamma^\rho_{\nu\lambda})\Gamma^\lambda_{\mu\sigma}} - \cancel{\Gamma^\rho_{\nu\lambda}(\delta\Gamma^\lambda_{\mu\sigma})}$$
$$= \nabla_\mu \delta\Gamma^\rho_{\nu\sigma} - \nabla_\nu \delta\Gamma^\rho_{\mu\sigma}$$

$$\Rightarrow \delta R^\rho{}_{\sigma\mu\nu} = \nabla_\mu \delta\Gamma^\rho_{\nu\sigma} - \nabla_\nu \delta\Gamma^\rho_{\mu\sigma}$$

$$\Rightarrow \delta R_{\sigma\nu} = \delta R^\rho{}_{\sigma\rho\nu} = \nabla_\rho \delta\Gamma^\rho_{\sigma\nu} - \nabla_\nu \delta\Gamma^\rho_{\rho\nu}$$

$$\Rightarrow \delta R = \delta(g^{\mu\nu}R_{\mu\nu})$$
$$= (\delta g^{\mu\nu})R_{\mu\nu} + g^{\mu\nu}\delta R_{\mu\nu}$$
$$= R_{\mu\nu}\,\delta g^{\mu\nu} + g^{\mu\nu}\nabla_\rho \delta\Gamma^\rho_{\mu\nu} - g^{\mu\nu}\nabla_\nu \delta\Gamma^\sigma_{\sigma\mu}$$

$\nabla(g\otimes T) = \underbrace{\nabla g \otimes T}_{0} + g\otimes\nabla T$

$\nabla g = 0$

$$= R_{\mu\nu}\delta g^{\mu\nu} + \nabla_\rho(g^{\mu\nu}\delta\Gamma^\rho_{\mu\nu}) - g^{\mu\nu}g^\rho{}_\nu\nabla_\rho\delta\Gamma^\sigma_{\sigma\mu}$$

$$= R_{\mu\nu}\delta g^{\mu\nu} + \nabla_\rho\left(g^{\mu\nu}\delta\Gamma^\rho_{\mu\nu} - g^{\mu\nu}g^\rho{}_\nu\delta\Gamma^\sigma_{\sigma\mu}\right)$$

Now $0 = \delta S$

$$= \int_M \delta(R\sqrt{-g})\, dx^0\wedge\cdots\wedge dx^n$$

$$= \int_M \left(R\,\delta(\sqrt{-g}) + \sqrt{-g}\,\delta R\right) dx^0\wedge\cdots\wedge dx^n$$

$$= \int_M R\left(-\tfrac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}\right) + \sqrt{-g}\left(R_{\mu\nu}\delta g^{\mu\nu} + \nabla_\rho(g^{\mu\nu}\delta\Gamma^\rho_{\mu\nu} - g^{\mu\nu}\delta\Gamma^\sigma_{\sigma\mu})\right) dx^0\wedge\cdots\wedge dx^n$$

$$= \int_M \left(R_{\mu\nu} - \tfrac{1}{2}Rg_{\mu\nu}\right)\delta g^{\mu\nu}\sqrt{-g}\, dx^0\wedge\cdots\wedge dx^n + \int_M \nabla_\rho(g^{\mu\nu}\delta\Gamma^\rho_{\mu\nu} - g^{\mu\nu}\delta\Gamma^\sigma_{\sigma\mu})\sqrt{-g}\, dx^0\wedge\cdots\wedge dx^n$$

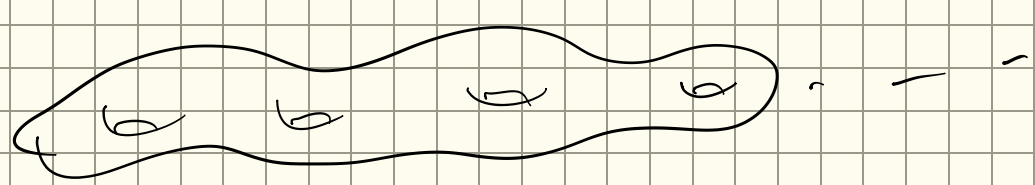
$\Rightarrow R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 0$, $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}$

let us contract : $g^{\mu\nu}R_{\mu\nu} - \frac{1}{2}Rg^{\mu\nu}g_{\mu\nu} = 0$

$$\Rightarrow R - \tfrac{1}{2}Rn = 0$$

$$\Rightarrow R\left(1 - \tfrac{1}{2}n\right) = 0$$

$\Rightarrow R = 0$ or $n = 2$



consider $S(g) = \int (R - 2\Lambda)\, dV_g$

$\Rightarrow R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 0$

$\Rightarrow G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$

To add matter consider $S(g) = \int (R - 2\Lambda + \mathcal{L}_{matter})\, dV_g$

$$\delta(\mathcal{L}_{matter}\sqrt{-g}) = \delta(\sqrt{-g})\mathcal{L}_{matter} + \sqrt{-g}\,\delta\mathcal{L}_{matter}$$
$$= -\frac{1}{2}\sqrt{-g}\,T_{\mu\nu}\,\delta g^{\mu\nu}$$

Where $T_{\mu\nu} = g_{\mu\nu}\mathcal{L}_{matter} - 2\frac{\delta\mathcal{L}_{matter}}{\delta g^{\mu\nu}}$

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = T_{\mu\nu}}$$

$\nabla_\mu T^{\mu\nu} = 0 \leftarrow$ Emmy Noether